EST II — MATH LEVEL 2 • UNIT 8.01

C Trigonometric Laws, Polar Coordinates & Conic Sections

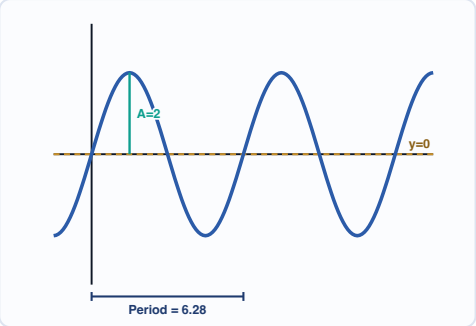
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Trigonometric Laws, Polar Coordinates & Conic Sections Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?



A $2 + 3i$

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

2 MCQ

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

A 0

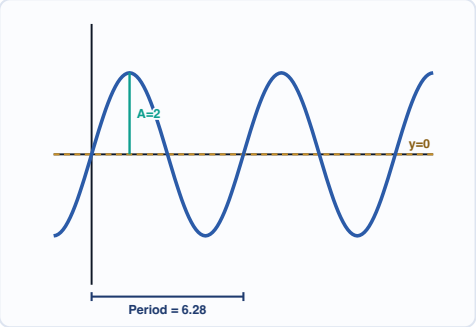
B 2

C -1

D 3

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



A (1, 0) and (3, 0)

B (0, 1) and (0, 3)

C (-1, 0) and (-3, 0)

D (1, 0) and (-3, 0)

4 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

A [3, 4]

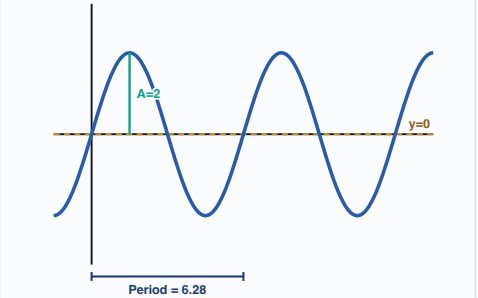
B [5, 8]

C [-3, -4]

D [4, 3]

5 MCQ

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?



A $\frac{3}{5}$

B $\frac{2}{5}$

C $\frac{1}{2}$

D $\frac{3}{4}$

6 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

A 14

B 17

C 8

D 11

7 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

A 96

B 192

C 98

D -96

8 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

A 0

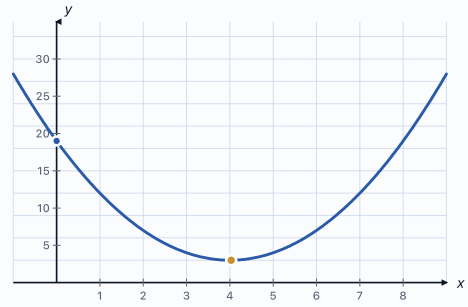
B -2

C 2

D 1

9 MCQ

In Trigonometry, Polar & Conic Sections (problem set #925), consider the function $f_{925}(x) = x^2 - 8x + 19$ illustrated on the coordinate axes. Express $f_{925}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{925}(6)$.



A 14

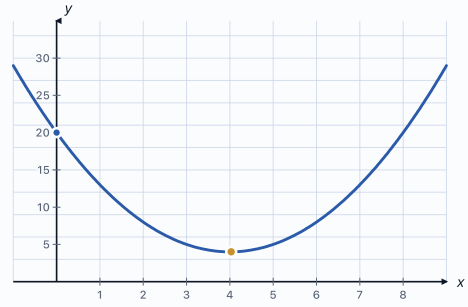
B 7

C 10

D 8

10 MCQ

Considering Trigonometry, Polar & Conic Sections (problem set #926), consider the function $f_{926}(x) = x^2 - 8x + 20$ illustrated on the coordinate axes. Express $f_{926}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{926}(7)$.



A 21

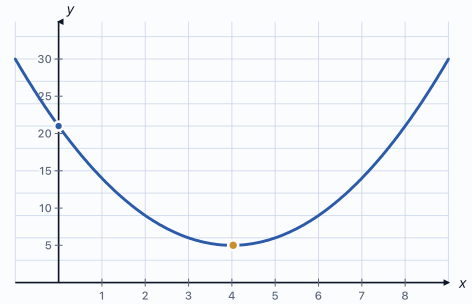
B 13

C 14

D 17

11 MCQ

Examining Trigonometry, Polar & Conic Sections (problem set #927), consider the function $f_{927}(x) = x^2 - 8x + 21$ illustrated on the coordinate axes. Express $f_{927}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{927}(8)$.



A 21

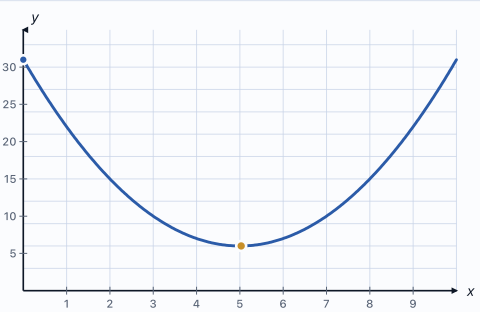
B 22

C 30

D 26

12 MCQ

For the configuration in Trigonometry, Polar & Conic Sections (problem set #928), consider the function $f_{928}(x) = x^2 - 10x + 31$ illustrated on the coordinate axes. Express $f_{928}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{928}(10)$.



A 32

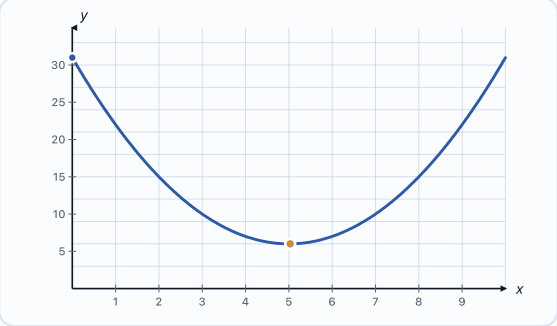
B 42

C 31

D 37

13 MCQ

Analyzing Trigonometry, Polar & Conic Sections (problem set #929), consider the function $f_{929}(x) = x^2 - 10x + 32$ illustrated on the coordinate axes. Express $f_{929}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{929}(11)$.



A 50

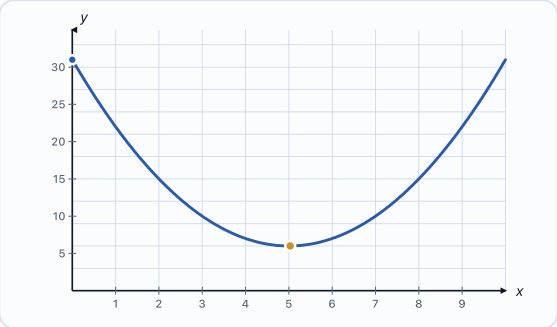
B 44

C 55

D 43

14 MCQ

Within Trigonometry, Polar & Conic Sections (problem set #930), consider the function $f_{930}(x) = x^2 - 10x + 33$ illustrated on the coordinate axes. Express $f_{930}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{930}(12)$.



A 70

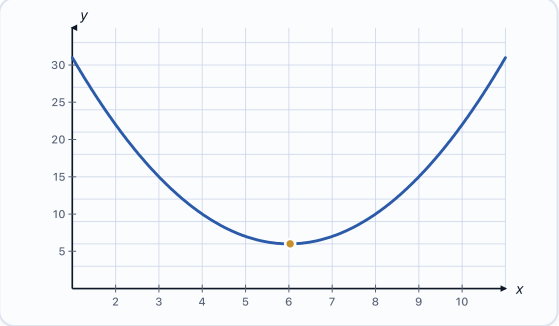
B 57

C 65

D 58

15 MCQ

Considering Trigonometry, Polar & Conic Sections (problem set #931), consider the function $f_{931}(x) = x^2 - 12x + 45$ illustrated on the coordinate axes. Express $f_{931}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{931}(14)$.



A 88

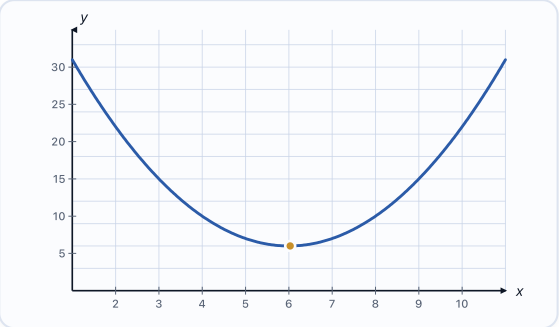
B 73

C 82

D 74

16 MCQ

Examining Trigonometry, Polar & Conic Sections (problem set #932), consider the function $f_{932}(x) = x^2 - 12x + 46$ illustrated on the coordinate axes. Express $f_{932}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{932}(15)$.



A 101

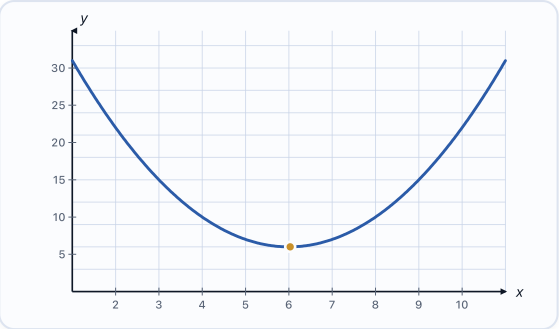
B 91

C 92

D 107

17 MCQ

For the configuration in Trigonometry, Polar & Conic Sections (problem set #933), consider the function $f_{933}(x) = x^2 - 12x + 47$ illustrated on the coordinate axes. Express $f_{933}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{933}(16)$.



A 128

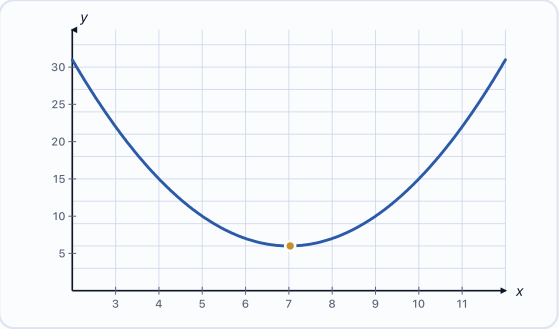
B 111

C 122

D 112

18 MCQ

Analyzing Trigonometry, Polar & Conic Sections (problem set #934), consider the function $f_{934}(x) = x^2 - 14x + 61$ illustrated on the coordinate axes. Express $f_{934}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{934}(9)$.



A 35

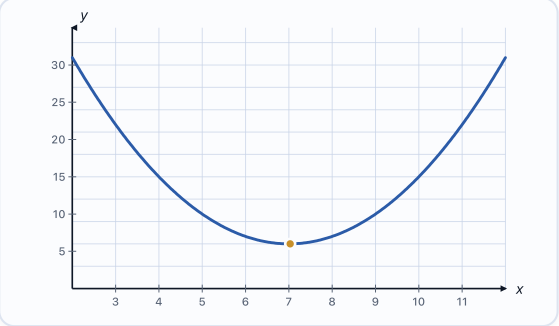
B 16

C 28

D 26

19 MCQ

Within Trigonometry, Polar & Conic Sections (problem set #935), consider the function $f_{935}(x) = x^2 - 14x + 62$ illustrated on the coordinate axes. Express $f_{935}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{935}(10)$.



A 35

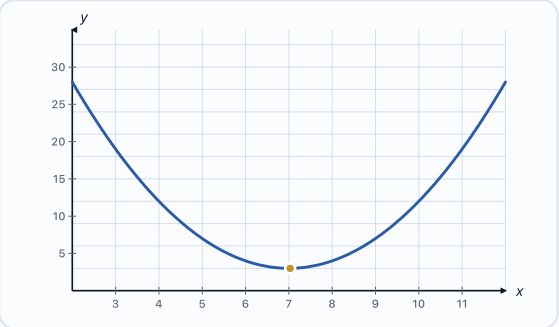
B 42

C 22

D 32

20 MCQ

Considering Trigonometry, Polar & Conic Sections (problem set #936), consider the function $f_{936}(x) = x^2 - 14x + 52$ illustrated on the coordinate axes. Express $f_{936}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{936}(11)$.



A 22

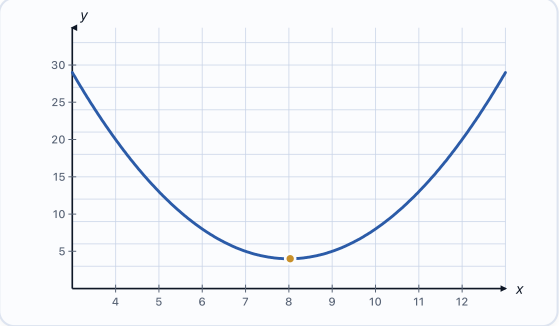
B 18

C 29

D 19

21 MCQ

Examining Trigonometry, Polar & Conic Sections (problem set #937), consider the function $f_{937}(x) = x^2 - 16x + 68$ illustrated on the coordinate axes. Express $f_{937}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{937}(13)$.



A 41

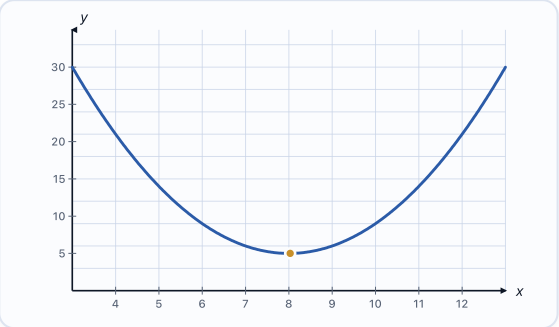
B 29

C 33

D 28

22 MCQ

For the configuration in Trigonometry, Polar & Conic Sections (problem set #938), consider the function $f_{938}(x) = x^2 - 16x + 69$ illustrated on the coordinate axes. Express $f_{938}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{938}(14)$.



A 40

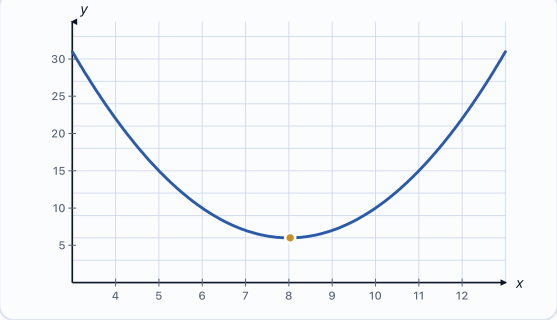
B 46

C 54

D 41

23 MCQ

Analyzing Trigonometry, Polar & Conic Sections (problem set #939), consider the function $f_{939}(x) = x^2 - 16x + 70$ illustrated on the coordinate axes. Express $f_{939}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{939}(15)$.



A 61

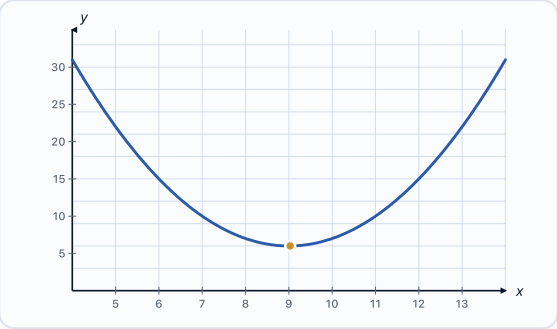
B 69

C 55

D 54

24 MCQ

Within Trigonometry, Polar & Conic Sections (problem set #940), consider the function $f_{940}(x) = x^2 - 18x + 88$ illustrated on the coordinate axes. Express $f_{940}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{940}(17)$.



A 87

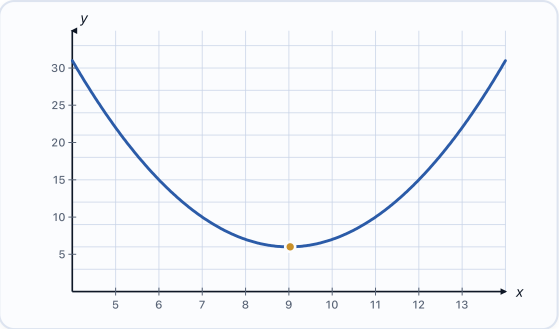
B 71

C 70

D 78

25 MCQ

Considering Trigonometry, Polar & Conic Sections (problem set #941), consider the function $f_{941}(x) = x^2 - 18x + 89$ illustrated on the coordinate axes. Express $f_{941}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{941}(18)$.



A 106

B 89

C 97

D 88